

CHEM 101 Topic 7: Chemical Reactions — Problem Set

Instructions: Complete the following problems using the analysis method practiced in your guided assignment. Show all your work, including your balancing steps, reaction type classification, and solubility rule justifications for any precipitates.

Section 1: Writing and Balancing Equations

Problem 1: Identify the reactants and products in the following equation: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$

Problem 2: Balance the following equation: $\text{Al} + \text{CuCl}_2 \rightarrow \text{AlCl}_3 + \text{Cu}$

Problem 3: Balance the following equation: $\text{C}_2\text{H}_6 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

Section 2: Types of Chemical Reactions

Problem 4: Classify the reaction: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$

Problem 5: Predict the products for the combustion of propane (C_3H_8). Write the unbalanced equation.

Problem 6: Classify the following two reactions as either Single Replacement or Double Replacement: (a) $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$; (b) $\text{BaCl}_2 + \text{Na}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{NaCl}$

Section 3: Solubility Rules & Precipitation

Problem 7: Will a precipitate form if you mix aqueous NaNO_3 and aqueous KCl ? Justify your answer using solubility rules.

Problem 8: Predict the precipitate that forms when aqueous lead(II) nitrate, $\text{Pb}(\text{NO}_3)_2$, is mixed with aqueous sodium chloride, NaCl .

Problem 9: Predict and balance the reaction between aqueous silver nitrate (AgNO_3) and aqueous potassium phosphate (K_3PO_4).

Section 4: Clinical Applications & Synthesis

Problem 10: Antacids often use magnesium hydroxide, $\text{Mg}(\text{OH})_2$, to neutralize stomach acid (HCl). What type of reaction is this?

Problem 11: A common component of kidney stones is calcium phosphate. Write the chemical formula for calcium phosphate given Ca^{2+} and PO_4^{3-} .

Problem 12: Write the unbalanced equation for the combustion of ethanol ($\text{C}_2\text{H}_5\text{OH}$).